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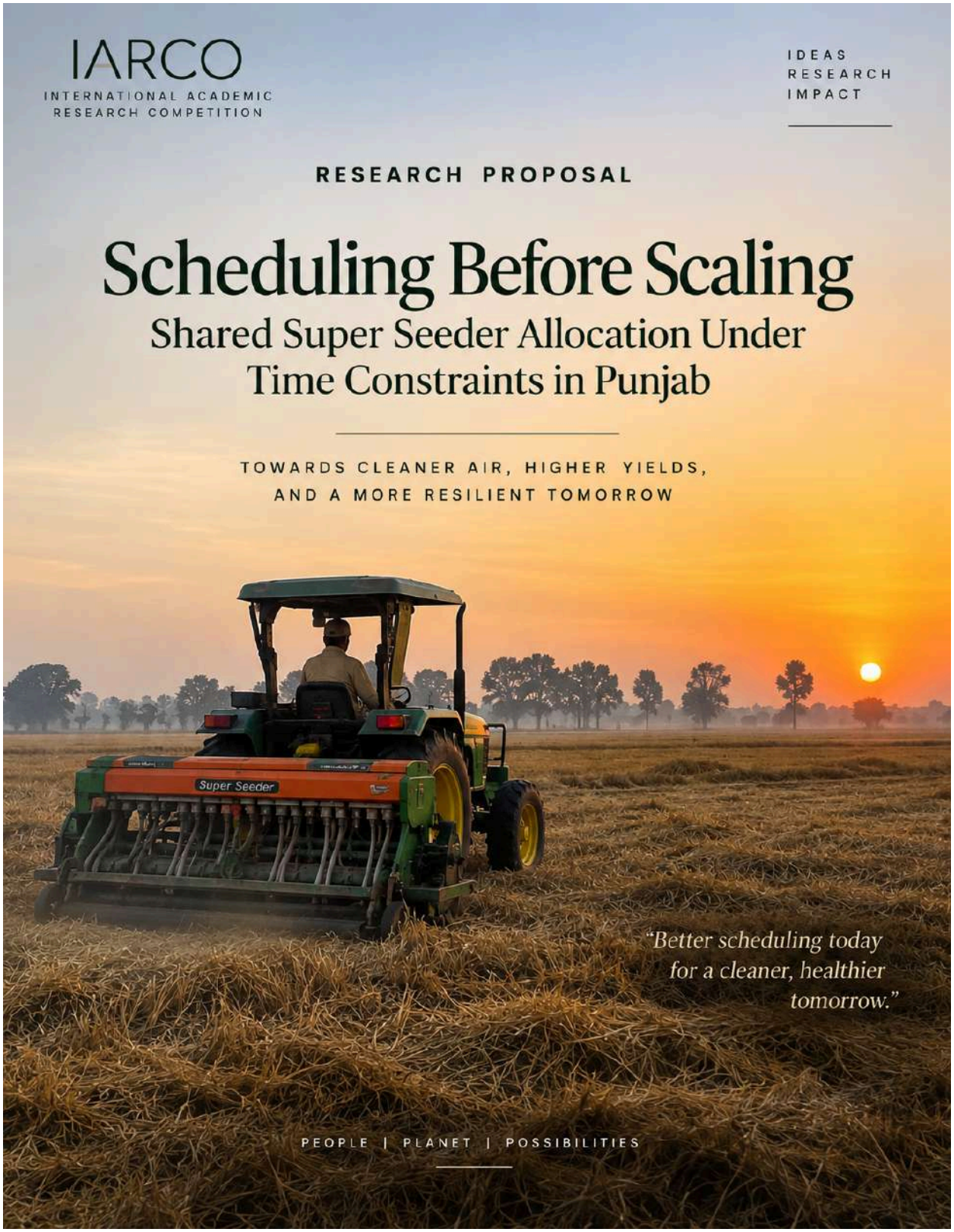
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Research Topic: Operations Research and Agricultural Resource Allocation

Title: *Scheduling Before Scaling: Shared Super Seeder Allocation Under Time Constraints in Punjab*


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Scheduling Before Scaling
Shared Super Seeder Allocation Under
Time Constraints in Punjab
TOWARDS CLEANER AIR, HIGHER YIELDS,
AND A MORE RESILIENT TOMORROW
Super Seeder
"Better scheduling today
for a cleaner, healthier
tomorrow."
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Title: Scheduling Before Scaling: Shared Super Seeder Allocation Under Time Constraints in Punjab

Research Problem

The crop-residue management problem of Punjab occurs within a tight agricultural calendar. After the paddy harvesting, farmers need to manage their remaining residue and prepare the fields for the sowing of wheat. One machine that is used for this is the Super Seeder. It cuts and incorporates paddy residue while also sowing wheat. Punjab Agricultural University (PAU) reports a field capacity of approximately 0.22–0.27 hectares per hour for the machine [4]. Timing is important because PAU recommends the first fortnight of November for wheat sowing and advises farmers to complete sowing by around 15 November. It also reports an average wheat-yield loss of approximately 1.5 quintals per acre for each week of delay [5].

10 Punjab has invested substantially in Crop Residue Management (CRM) machinery and Custom Hiring Centres (CHCs), which allow farmers to access machinery without individually purchasing it [6]. Yet having machines in the system does not necessarily mean that every farmer can access one when required. In a 2025 study of 65 CHCs across Ludhiana, Patiala and Sangrur, Jain, Kemanth and Sarao found that most surveyed centres rented machinery on a First-Come, First-Served (FCFS) basis [1]. Earlier farmer-level research by Kemanth, Singh and Ignatious also identified timely access to CRM machinery as a practical difficulty [2].

These findings raise a specific operational question. Suppose several farms require the same limited machinery during overlapping periods. FCFS gives priority according to arrival order, regardless of how much time each farmer has left before service becomes too late. A deadline-aware system would instead take the remaining service window into account.

This study will test whether that difference matters. Shared Super Seeder access will be represented as a capacity-constrained scheduling problem, and FCFS will be compared with a deadline-aware allocation rule. Machinery capacity will then be varied separately. This allows the study to measure how much improvement comes from changing the order of service and how much comes from adding machines.

Existing Literature

Research on CRM in Punjab provides evidence from both farmers and machinery providers. Kemanth, Singh and Ignatious surveyed farmers across 11 Punjab districts and identified timely machinery access among the practical barriers surrounding CRM adoption [2]. Jain, Kemanth and Sarao subsequently studied 65 CHCs in three Punjab districts, examining machinery availability, demand, rental practices and utilisation. Their report states that most surveyed CHCs rent machines on an FCFS basis [1]. These studies establish the local context for the proposed research: machinery is shared, access is time-sensitive, and FCFS is a relevant real-world baseline.

A study particularly close to the proposed research is Farahani et al. [3], who examine limited Happy Seeder availability in the northwestern Indian rice–wheat system. Their model accounts for waiting for machinery and the possibility that farmers choose to burn residue. The scheduling component, however, serves a different research question. The government in their model already knows the Happy Seeder schedule , and the researchers investigate what information about that schedule should be disclosed to farmers [3]. The present study instead focuses on the rule used to determine the service order.

Agricultural machinery scheduling has also been studied more generally. Wang and Huang [8] develop a framework for dispatching shared agricultural machinery when individual farms have time windows. Their analysis uses more than 400 machinery requests and incorporates factors including processing requirements and travel between farms. Guo et al. [9] also investigate agricultural machinery scheduling under time-window constraints. This literature shows that time-constrained machinery allocation is already an established operations-research problem.

The proposed study therefore does not depend on claiming that agricultural scheduling itself is new. The gap is narrower. Among the literature reviewed for this proposal, I did not identify a study that directly compares the FCFS practice documented among Punjab CHCs with deadline-aware Super Seeder allocation and then compares the resulting scheduling gain with the effect of increasing machinery capacity. This specific comparison forms the research gap addressed in the proposed study.

Research Question

The central research question is:

> To what extent can deadline-aware scheduling reduce missed service deadlines in shared Super Seeder systems compared with First-Come, First-Served allocation in Punjab, and how does this effect change with machinery capacity?

Three supporting questions will be considered:

1. How does deadline-aware scheduling change the proportion of farms receiving Super Seeder service within their required service window compared with FCFS?
2. How does the difference between FCFS and deadline-aware scheduling change as machinery availability and the concentration of demand vary?
3. Can deadline-aware scheduling with a given number of machines produce service performance comparable to FCFS with a larger fleet, and under what simulated conditions?

The study will measure machinery-service outcomes rather than directly estimating changes in crop-residue burning.

Methodology

The research will use a discrete-event simulation developed in Python. The model will represent a group of farms requesting service from a limited number of shared Super Seeders. Simulation is suitable for the research question because the scheduling rules can be tested under identical demand conditions without changing real farmers' access to machinery.

Each simulated farm i will have a cultivated area A_i, a time at which it becomes ready for service, and a deadline representing the end of its modeled service window. Farm-size inputs will be Calibrated using published Punjab agricultural statistics [7]. The model will not assume that every farm requires the machine for the same amount of time.

Parameter	Treatment in simulation
Farm area A_i	Calibrated using published Punjab agricultural statistics [7]
Readiness time a_i	Generated according to the demand-concentration scenario
Service deadline d_i	Assigned from the modeled service-window scenario
Field capacity q	0.22–0.27 ha/hour [4]
Service time s_i	$\frac{A_i}{q}$
Fleet size M	Varied experimentally

PAU reports a Super Seeder field capacity of 0.22–0.27 ha/hour [4]. If q represents field capacity, the service time required for farm i will be calculated as:

$$s_i = \frac{A_i}{q}$$

The reported PAU range will also allow machine productivity to be varied during sensitivity testing rather than relying on one assumed value.

Service windows will be constructed around the time-sensitive transition between paddy harvesting and wheat sowing. PAU's guidance on the preferred wheat-sowing period will inform plausible timing assumptions [5]. Parameters for which reliable Punjab data cannot be obtained will be stated as simulation assumptions and varied across scenarios.

Two allocation rules will initially be tested. The baseline will be First-Come, First-Served, reflecting the practice reported by Jain et al. [1]. Under FCFS, when a machine becomes available, the farm that entered the queue first receives service.

The alternative will be Earliest-Deadline-First (EDF). Under EDF, the available machine is assigned to the waiting farm whose service deadline occurs soonest. If two waiting farms have the same deadline, the farm that entered the queue first will receive priority. The two rules therefore differ in the information used to establish priority: FCFS uses arrival order, while EDF uses remaining time before the service window closes.

For every direct comparison, both scheduling rules will receive the same simulated farms, farm areas, readiness times, deadlines and machinery capacity. Only the allocation rule will change. This is necessary to prevent differences in the generated farm population from being mistaken for scheduling effects.

Machinery capacity will then be varied. For example, the same demand scenario can be simulated using progressively larger fleets. Demand concentration will also be varied by changing the period over which farms become ready for service. In concentrated-demand scenarios, farm readiness times will fall within a shorter period, creating greater overlap between machinery requests. In dispersed-demand scenarios, the same number of requests will be distributed across a wider period. This will allow the study to test whether the relative performance of FCFS and EDF changes as pressure on the shared machinery system increases.

Each scenario will be run repeatedly using independently generated farm populations. Simulation means will be monitored for convergence, and results will be reported with 95% confidence intervals so that differences between scheduling rules are not inferred from a single random realization.

The main outcome will be the Deadline Miss Rate (DMR):

$$DMR = \frac{Number\ of\ farms\ missing\ their\ service\ deadline}{Total\ number\ of\ farms\ requesting\ service}$$

Average waiting time and machinery utilisation will be recorded as secondary outcomes.For each scenario, mean DMR under FCFS and EDF will be compared together with its simulation uncertainty.

The analysis will also examine how the difference $DMR_{FCFS} - DMR_{EDF}$ changes with fleet size and demand concentration.

The comparison will then be made at two levels. First, DMR under FCFS and EDF will be compared at the same machinery capacity. Second, the effect of changing the scheduling rule will be compared with the effect of increasing the number of machines. One question, for example, is whether EDF using m machines can reach approximately the same DMR as FCFS using $m+1$ machines.

Several simplifications will be stated explicitly. Weather, breakdowns, tractor availability, detailed routing between farms, rental prices and informal machinery sharing may not be represented in the initial model. Wang and Huang's work shows that routing can itself affect shared-machinery scheduling [8], so excluding geography will be treated as a limitation rather than assumed to have no effect.

A missed service deadline will also not be classified as a case of stubble burning. Farahani et al. demonstrate that machinery waiting can enter farmers' residue-management decisions [3], but those decisions depend on other factors beyond the proposed simulation. The dependent variable in this study is timely machinery service.

Research Topic

The project sits between agricultural policy and operations research. Punjab's CRM programme has invested in machinery acquisition and shared machinery systems [6]. If farmers still experience difficulty obtaining machines during short periods of peak demand, one possible response is to increase the number of machines available. That is a capacity intervention.

There is another variable that can be tested separately: allocation. FCFS is simple to operate, but it does not use information about how close a farm is to the end of its service window. The simulation will measure whether incorporating that information produces a meaningful difference under the same machinery constraint.

This distinction gives the study practical relevance without requiring a predetermined result. A large difference between FCFS and EDF would suggest that service rules deserve more attention when evaluating machinery access. A small difference would indicate that changing queue order does little to relieve the constraint and that additional capacity, or other factors omitted from the model, may matter more. The results could therefore be useful at two levels. CHC operators could use the comparison to assess whether request priority should account for service deadlines, while policymakers could use the capacity comparison to judge whether improved allocation is likely to provide meaningful gains before additional machinery is funded. The study does not assume that either response will be superior; that is what the simulation is designed to test.

The methodological idea of agricultural machinery scheduling is already established [8], [9]. The specific contribution proposed here is the comparison of an observed Punjab allocation practice with deadline-aware scheduling, followed by a direct comparison between scheduling gains and capacity gains in a Super Seeder system.

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